Lecture 7: Eigenvalue Decomposition

Through a Photo-Filter Problem

AI 501: Mathematics for Artificial Intelligence

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Abstract

A square matrix can be understood by the directions it scales without turning. These notes develop eigenvalues and eigenvectors from that idea and show that the eigen-decomposition reduces any such matrix to a change of basis, a pure scaling, and a change back. A single geometric example, a diagonal-stretch photo filter, is carried throughout to show how this one factorisation explains a matrix's effect on area, its invertibility, and its repeated application, with symmetric matrices emerging as the best-behaved case.

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1 One problem: a diagonal-stretch photo filter

1.1 The filter

A photo-editing program has a filter that stretches a photo along its main diagonal. A photo is a grid of pixels, and each pixel sits at a position $x = (x_1, x_2)^T$ measured from the centre of the picture. The filter moves every pixel to a new position:

$$x_{\text{new}} = S\,x, \qquad S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

To see what this does, we test it on a simple image: a round logo of radius 1 (the unit circle) on a square canvas with corners at $(\pm 1, \pm 1)$. Figure 1 shows the result. The round logo becomes a tilted oval, and the square canvas becomes a slanted parallelogram.

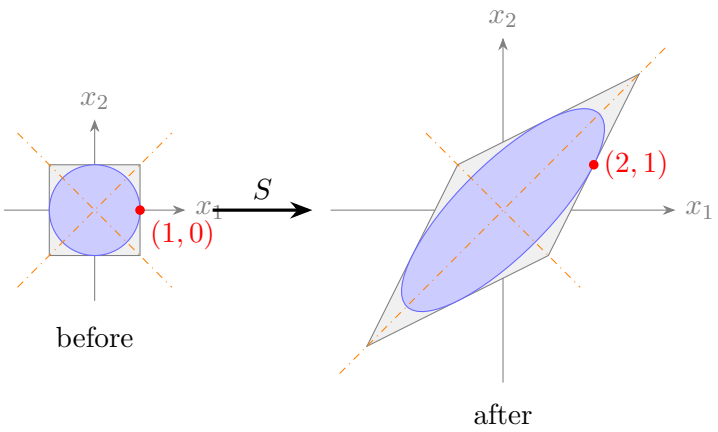


Figure 1: The diagonal-stretch filter S applied to a test image: a round logo on a square canvas. The red pixel at $(1, 0)$ moves to $(2, 1)$. The dash-dotted lines are the diagonal and the anti-diagonal; keep an eye on them.

1.2 The questions

Looking at Figure 1, four questions come up naturally.

1. Most pixels are moved *and* tilted. Are there directions the filter does not tilt? What does it do to them? (Sections 2–3)
2. Is there a simpler way to describe, or program, what S does? (Sections 5–6)

3. How much bigger is the picture afterwards? How do we build an *undo* button? What happens if the filter is applied five times? Can every filter be undone? (Section 7)
4. Why is this filter so well behaved, and what does a badly behaved one look like? (Section 8)

The answers to all four come from the eigenvalues and eigenvectors of S.

2 Eigenvectors and eigenvalues

2.1 The intuition: directions the filter does not tilt

Track three pixels through the filter:

$$S\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}2\\1\end{bmatrix}, \qquad S\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\3\end{bmatrix} = 3\begin{bmatrix}1\\1\end{bmatrix}, \qquad S\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}1\\-1\end{bmatrix} = 1\begin{bmatrix}1\\-1\end{bmatrix}.$$

The pixel at $(1, 0)$ is pushed off its line: it was on the horizontal axis and ends up pointing somewhere else. The pixel at $(1, 1)$ stays on the diagonal and moves three times further out. The pixel at $(1, -1)$ does not move at all. These two special directions are exactly the dash-dotted lines in Figure 1: the filter stretches the diagonal by 3 and leaves the anti-diagonal alone. Figure 2 shows the three pixels.

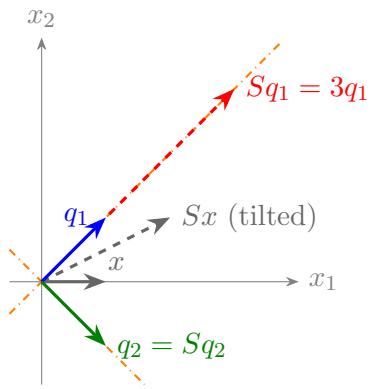


Figure 2: Three pixels under the filter. $x = (1, 0)^T$ is tilted to $(2, 1)^T$; $q_1 = (1, 1)^T$ stays on its line and is stretched by 3; $q_2 = (1, -1)^T$ stays exactly where it is (stretched by 1). q_1 and q_2 are eigenvectors; 3 and 1 are their eigenvalues.

2.2 Definition

For a square matrix $A \in \mathbf{R}^{n\times n}$, a vector $q \in \mathbf{R}^n$ with $q \neq 0$ is called an *eigenvector* of A if

$$\boxed{A\,q = \lambda\,q} \qquad \text{(the eigenvalue equation)}$$

for some number λ. The number λ is the *eigenvalue* of A associated with the eigenvector q. Together, the eigenvectors and eigenvalues make up the *eigen-decomposition* of A, and they analyse the structure of the matrix. For our filter, $q_1 = (1, 1)^T$ with $\lambda_1 = 3$ and $q_2 = (1, -1)^T$ with $\lambda_2 = 1$.

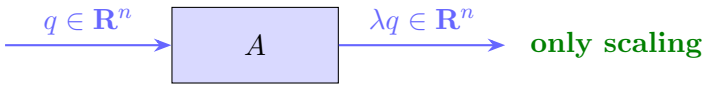
Two remarks on the definition.

- The condition $q \neq 0$ matters. The centre pixel 0 never moves ($S0 = 0 = \lambda 0$ for every λ), so allowing it would tell us nothing.
- Eigenvectors are only defined up to scale. If $Aq = \lambda q$, then $A(cq) = cAq = \lambda(cq)$ for any $c \neq 0$. *Every* pixel on the diagonal is stretched by 3, not just $(1, 1)$. What matters is the *direction* (the line through q), so we usually pick the representative of length one (we *normalize* it): $q_1 = \frac{1}{\sqrt{2}}(1, 1)^T$.

Eigenvalues and eigenvectors are only defined for *square* matrices: Aq and q must live in the same space (here: the same canvas) for $Aq = \lambda q$ to make sense.

2.3 The linear-transformation picture

In the block-diagram view the eigenvalue equation reads:



Geometrically, an eigenvector spans a line through the origin that A maps onto itself. Pixels on that line are stretched by λ (if $|\lambda| > 1$), shrunk (if $|\lambda| < 1$), flipped to the other side of the centre (if $\lambda < 0$), or squashed into the centre (if $\lambda = 0$), but never moved off the line.

Check yourself. Is the pixel direction $(0, 1)^T$ an eigenvector of S? Is $(-2, -2)^T$?
Answer. $S(0, 1)^T = (1, 2)^T$, which is not a multiple of $(0, 1)^T$, so no: the vertical axis gets tilted. $S(-2, -2)^T = (-6, -6)^T = 3(-2, -2)^T$, so yes: it lies on the diagonal, with eigenvalue 3.

3 Computing eigenvalues and eigenvectors

Move everything to one side, and write $\lambda q = \lambda I q$ so that both terms are a matrix times q:

$$Aq = \lambda q \quad \Longrightarrow \quad Aq - \lambda I q = 0 \quad \Longrightarrow \quad (A - \lambda I)\, q = 0.$$

We want a *nonzero* q that the matrix $A - \lambda I$ sends to zero. If $A - \lambda I$ were invertible, the only solution would be $q = (A - \lambda I)^{-1} 0 = 0$. So a nonzero solution exists exactly when $A - \lambda I$ is *not* invertible, that is, when its determinant is zero:

$$\boxed{p(\lambda) = \det(A - \lambda I) = 0}$$

The function $p(\lambda)$ is the *characteristic polynomial* of A. Expanding the determinant of an $n \times n$ matrix with λ on the diagonal gives a polynomial of degree n in λ.

(The slides write this as $\det(Aq - \lambda q) = 0$ as a shorthand; the determinant is of the *matrix* $A - \lambda I$, as in the lecture's tutorial problem.)

3.1 Vocabulary

A polynomial of degree n has n roots (counted with repetition), so:

- A has n eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, and correspondingly n eigenvectors $q_1, q_2, \ldots, q_n \in \mathbf{R}^n$.
- The *eigenspectrum* (or spectrum) of A is the set of its eigenvalues. For the filter: $\{3, 1\}$.
- The *eigenspace* is the span of the eigenvectors (for a given eigenvalue λ: all vectors q with $(A - \lambda I)q = 0$, the null space of $A - \lambda I$). For the filter, the eigenspace of $\lambda = 3$ is the whole diagonal line.

3.2 The recipe

- **Subtract λ from the diagonal.** Write down $A - \lambda I$; leave every off-diagonal entry unchanged.
- **Take the determinant.** This gives $p(\lambda)$, a polynomial in λ. Shortcut for 2×2: $p(\lambda) = \lambda^2 - (a+d)\lambda + (ad - bc)$.
- **Solve $p(\lambda) = 0$.** The roots are the eigenvalues. An $n \times n$ matrix gives n of them.
- **For each eigenvalue, solve $(A - \lambda I)q = 0$.** Plug in the number, write out one row as an equation, and express one entry of q in terms of the other. The other rows give the same equation, which is expected, so ignore them.
- **Pick a simple q.** Set one free entry to 1 (or any convenient number) to get a concrete vector.
- **Normalize.** Divide q by $\|q\| = \sqrt{q_1^2 + \cdots + q_n^2}$.
- **Check.** Compute Aq and confirm it equals λq.

Quick sanity checks: the eigenvalues add up to $a + d$ (the diagonal sum), and they multiply to $\det(A)$.

3.3 The recipe on the filter

Eigenvalues.

$$\det(S - \lambda I) = \det \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1),$$

so $\lambda_1 = 3$ and $\lambda_2 = 1$, the two stretch factors we spotted by eye.

Eigenvector for $\lambda_1 = 3$.

$$(S - 3I)v = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \quad \Longrightarrow \quad -v_1 + v_2 = 0 \quad \Longrightarrow \quad v_2 = v_1.$$

(The second row says the same thing, because we chose λ to make $S - 3I$ singular.) Every vector $v_1(1,1)^T$ works: the diagonal.

Eigenvector for $\lambda_2 = 1$.

$$(S - I)v = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \quad \Longrightarrow \quad v_1 + v_2 = 0 \quad \Longrightarrow \quad v_2 = -v_1.$$

Every vector $v_1(1,-1)^T$ works. We take the representative $(-1,1)^T$, which points in the same line (it is -1 times $(1,-1)^T$); the reason for this choice appears in Section 6.

Normalize. Both vectors have length $\sqrt{2}$:

$$q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \ (\lambda_1 = 3), \qquad q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \ (\lambda_2 = 1).$$

Check yourself. For a 2×2 matrix $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$, write $p(\lambda)$ and check it against the filter. *Answer.* $(a-\lambda)(d-\lambda) - bc = \lambda^2 - (a+d)\lambda + (ad - bc)$. For S: $a + d = 4$ and $ad - bc = 3$, giving $\lambda^2 - 4\lambda + 3$, as above.

4 Practice by hand: a filter with a negative eigenvalue

$$M = \begin{bmatrix} 3 & 2 \\ 2 & 0 \end{bmatrix}.$$

4.1 The eigenvalues

Subtract λ from the diagonal and take the determinant:

$$\begin{aligned}\det(M-\lambda I) = \det\begin{bmatrix}3-\lambda & 2\\ 2 & -\lambda\end{bmatrix} &= (3-\lambda)(-\lambda)-(2)(2)\\ &= (-3\lambda+\lambda^2)-4\\ &= \lambda^2-3\lambda-4\\ &= (\lambda-4)(\lambda+1).\end{aligned}$$

Setting this to zero gives $\lambda_1 = 4$ and $\lambda_2 = -1$.

Sanity check: $4+(-1) = 3 = 3+0$ (the diagonal sum) and $4\times(-1) = -4 = \det(M)$. ✓

4.2 The eigenvector for $\lambda_1 = 4$

Solve $(M-4I)v = 0$:

$$\begin{bmatrix}-1 & 2\\ 2 & -4\end{bmatrix}\begin{bmatrix}v_1\\ v_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix} \quad\Longrightarrow\quad \begin{cases}-v_1+2v_2=0\\ 2v_1-4v_2=0\end{cases}$$

The first equation gives $v_1 = 2v_2$. The second equation is -2 times the first, so it adds nothing. So $\mathbf{v}_1 = v_2(2,1)^T$. Its length is $\sqrt{(2v_2)^2+v_2^2} = v_2\sqrt{5}$, so choose $v_2 = 1/\sqrt{5}$:

$$q_1 = \frac{1}{\sqrt{5}}\begin{bmatrix}2\\ 1\end{bmatrix}.$$

4.3 The eigenvector for $\lambda_2 = -1$

Solve $(M+I)v = 0$ (subtracting -1 means adding 1 to the diagonal):

$$\begin{bmatrix}4 & 2\\ 2 & 1\end{bmatrix}\begin{bmatrix}v_1\\ v_2\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix} \quad\Longrightarrow\quad 2v_1+v_2=0 \quad\Longrightarrow\quad v_2 = -2v_1.$$

(The first row, $4v_1+2v_2=0$, is twice the second.) Taking $v_1 = 1$ gives $(1,-2)^T$, with length $\sqrt{5}$, so

$$q_2 = \frac{1}{\sqrt{5}}\begin{bmatrix}1\\ -2\end{bmatrix}.$$

Always verify. Ignoring the $1/\sqrt{5}$, which cancels:

$$M\begin{bmatrix}2\\ 1\end{bmatrix} = \begin{bmatrix}6+2\\ 4+0\end{bmatrix} = \begin{bmatrix}8\\ 4\end{bmatrix} = 4\begin{bmatrix}2\\ 1\end{bmatrix}, \qquad M\begin{bmatrix}1\\ -2\end{bmatrix} = \begin{bmatrix}3-4\\ 2+0\end{bmatrix} = \begin{bmatrix}-1\\ 2\end{bmatrix} = -1\begin{bmatrix}1\\ -2\end{bmatrix}.\checkmark$$

4.4 What this matrix does to the logo

As a filter, M stretches the direction $(2,1)$ by 4 and *flips* the direction $(1,-2)$: a pixel on that line jumps to the opposite side of the centre, at the same distance. The logo becomes an oval four times as long as it is wide, and it is also mirrored across the line through $(2,1)$ (Figure 3). A negative eigenvalue still keeps the direction on its own line; it just reverses which way along the line the pixel points. Notice also that q_1 and q_2 are perpendicular, $(2,1)\cdot(1,-2) = 0$, just like the diagonal and anti-diagonal of S. Both matrices are symmetric, and Section 8 explains why that matters.

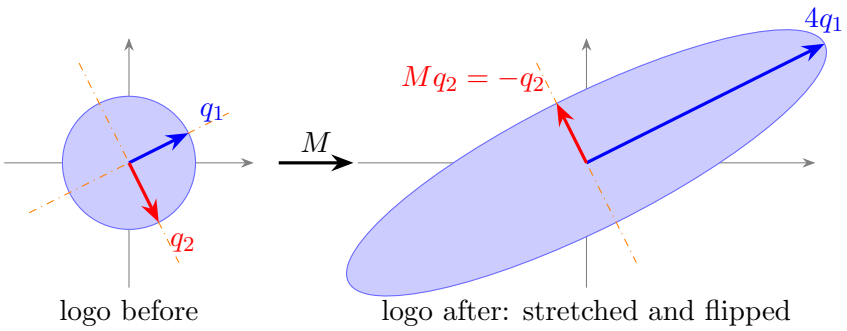


Figure 3: The practice filter $M = \left[\begin{smallmatrix} 3 & 2 \\ 2 & 0 \end{smallmatrix}\right]$. The direction q_1 is stretched by $\lambda_1 = 4$; the direction q_2 keeps its length but is flipped ($\lambda_2 = -1$), so q_2 now points the opposite way.

5 The eigen-decomposition $A = Q\Lambda Q^{-1}$

5.1 Stacking the eigenvalue equations

Write the n eigenvalue equations side by side, $Aq_1 = \lambda_1 q_1$, $Aq_2 = \lambda_2 q_2$, $\ldots$, $Aq_n = \lambda_n q_n$, and collect the eigenvectors as the columns of Q and the eigenvalues on the diagonal of Λ:

$$Q = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} (q_1)_1 & (q_2)_1 & \cdots & (q_n)_1 \\ (q_1)_2 & (q_2)_2 & \cdots & (q_n)_2 \\ \vdots & \vdots & \ddots & \vdots \\ (q_1)_n & (q_2)_n & \cdots & (q_n)_n \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Column i of AQ is Aq_i. Column i of $Q\Lambda$ is q_i multiplied by λ_i (a diagonal matrix on the right scales the columns). So all n equations at once read

$$AQ = Q\Lambda.$$

Multiplying on the right by Q^{-1} gives the *eigen-decomposition* (EVD):

$$\boxed{A = Q\,\Lambda\,Q^{-1}}$$

The last step needs Q to be invertible, that is, the n eigenvectors must be linearly independent. This holds for every matrix in these notes, and Section 8 shows it always holds for symmetric matrices. Shapes: Q, Λ and Q^{-1} are all $n \times n$, so $Q\Lambda Q^{-1}$ is $n \times n$, which matches A.

5.2 The filter's decomposition

With the eigenvectors from Section 3,

$$Q = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \qquad Q^{-1} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = Q^T.$$

(Check: $Q^TQ = \frac{1}{2}\left[\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right] = I$. Section 8 explains why the inverse is just the transpose.) Multiplying out,

$$Q\Lambda Q^{-1} = \frac{1}{2}\begin{bmatrix} 3 & -1 \\ 3 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = S. \checkmark$$

5.3 The practice filter's decomposition

Ordering the eigenvalues as $(4, -1)$,

$$Q = \frac{1}{\sqrt{5}}\begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix}, \qquad \Lambda = \begin{bmatrix} 4 & 0 \\ 0 & -1 \end{bmatrix}, \qquad Q^{-1} = Q^T = Q,$$

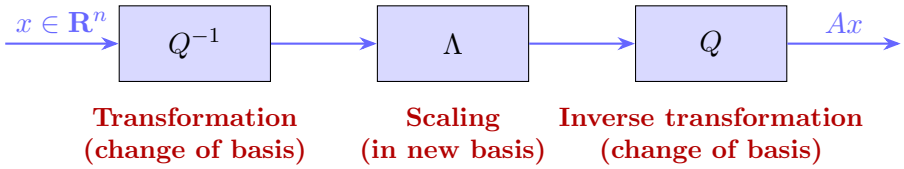
$$Q\Lambda Q^{-1} = \frac{1}{5}\begin{bmatrix}8 & -1\\ 4 & 2\end{bmatrix}\begin{bmatrix}2 & 1\\ 1 & -2\end{bmatrix} = \frac{1}{5}\begin{bmatrix}15 & 10\\ 10 & 0\end{bmatrix} = \begin{bmatrix}3 & 2\\ 2 & 0\end{bmatrix} = M.\checkmark$$

Check yourself. If we list the filter's eigenvalues in the order $(1, 3)$ instead, what changes? *Answer.* The diagonal of Λ and the columns of Q swap together, and $Q\Lambda Q^{-1}$ is still S. The order is a choice; the pairing of λ_i with q_i is not.

6 What the decomposition means: rotate, stretch, rotate back

6.1 Three simple steps instead of one complicated one

Read $Ax = Q\Lambda Q^{-1}x$ from right to left, one matrix at a time:



1. $Q^{-1}x$ rewrites x in the *eigenbasis*: it finds the coefficients c with $x = Qc = c_1q_1 + \cdots + c_nq_n$.
2. Λ scales coordinate i by λ_i, each coordinate on its own, with no mixing.
3. Q converts back to the original coordinates: $Q(\Lambda c) = \lambda_1 c_1 q_1 + \cdots + \lambda_n c_n q_n$.

Put together:

$$Ax = \sum_{i=1}^{n} \lambda_i\, c_i\, q_i \qquad \text{where } x = \sum_{i=1}^{n} c_i\, q_i.$$

Split the input along the eigenvectors, scale each piece by its eigenvalue, add up. A matrix that looks complicated in the standard basis is *only scaling* in its eigenbasis.

6.2 For the filter, the three steps are rotations and a stretch

The filter's $Q = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix}1 & -1\\ 1 & 1\end{smallmatrix}\right]$ is the matrix that rotates the canvas by 45° anticlockwise (it sends e_1 to the diagonal and e_2 to the anti-diagonal). This is why we chose $q_2 = \frac{1}{\sqrt{2}}(-1, 1)^T$: with that sign, Q is a pure rotation. Its inverse $Q^{-1} = Q^T$ rotates back by 45° clockwise. So

$$S = \underbrace{Q}_{\text{rotate } +45^\circ} \quad \underbrace{\begin{bmatrix}3 & 0\\ 0 & 1\end{bmatrix}}_{\text{stretch horizontally by 3}} \quad \underbrace{Q^T}_{\text{rotate } -45^\circ}.$$

To program the filter, a developer does not need to think about S at all: turn the photo so the diagonal lies flat, stretch it sideways by 3, and turn it back. The eigenvectors say *which way* to turn; the eigenvalues say *how much* to stretch.

Worked example. Send the red pixel $x = (1, 0)^T$ from Figure 1 through the three steps.

1. Rotate $-45°$: $c = Q^Tx = \frac{1}{\sqrt{2}}(1, -1)^T$. (In the eigenbasis, the pixel is $\frac{1}{\sqrt{2}}$ along the diagonal and $-\frac{1}{\sqrt{2}}$ along the anti-diagonal.)
2. Stretch: $\Lambda c = \frac{1}{\sqrt{2}}(3, -1)^T$. Only the diagonal part grows.
3. Rotate $+45°$: $Q\Lambda c = \frac{1}{2}\left[\begin{smallmatrix}1 & -1\\ 1 & 1\end{smallmatrix}\right](3, -1)^T = \frac{1}{2}(4, 2)^T = (2, 1)^T$.

This is Sx, the position we found in Figure 1. ✓

6.3 Visualization

Figure 4 follows the logo around the loop. Straight across the top, S turns the round logo into a tilted oval whose axes lie along q_1 and q_2. Going the long way round, Q^{-1} turns the eigenvectors onto the standard axes e_1, e_2, Λ stretches along those axes by $\lambda_1 = 3$ and $\lambda_2 = 1$, and Q turns the result back. Both routes give the same oval, which is what $S = Q\Lambda Q^{-1}$ says.

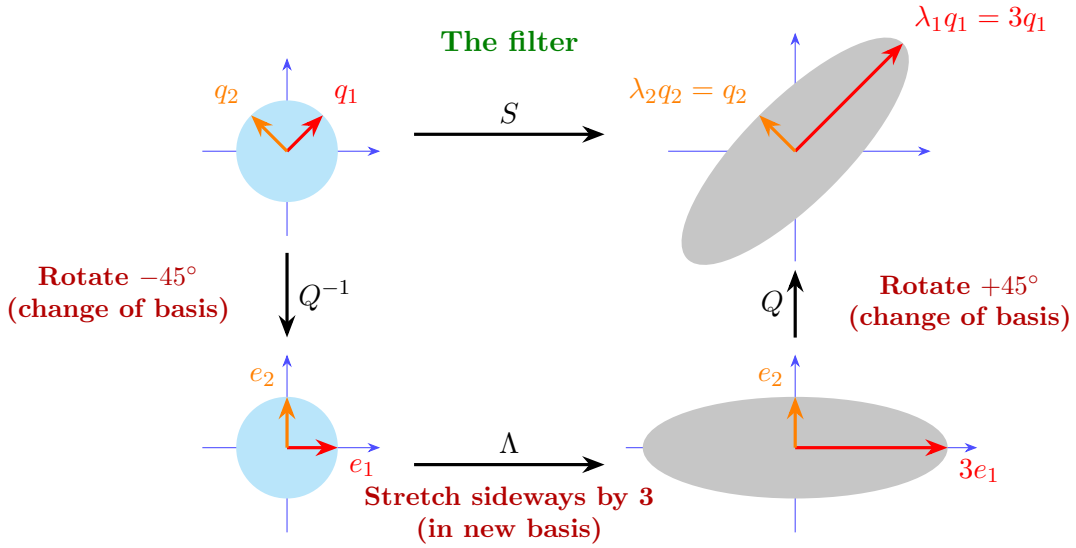


Figure 4: The filter S as a loop: rotate, stretch along the axes, rotate back. The oval's long axis lies along q_1 (stretched by 3) and its short axis along q_2 (unchanged).

Check yourself. A second filter stretches the diagonal by 2 and *shrinks* the anti-diagonal to half its length. Write it as $Q\Lambda Q^T$ and multiply out.
Answer. Same Q, $\Lambda = \mathrm{diag}(2, \frac{1}{2})$: $Q\Lambda Q^T = \frac{1}{2}\begin{bmatrix} 2 & -1/2 \\ 2 & 1/2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 5/2 & 3/2 \\ 3/2 & 5/2 \end{bmatrix} = \begin{bmatrix} 1.25 & 0.75 \\ 0.75 & 1.25 \end{bmatrix}$. Designing a filter is easy in the eigenbasis: pick the directions and the stretch factors.

7 Consequences: area, undo, repeat, flatten

7.1 Determinant: how much bigger is the picture?

Using $\det(BC) = \det(B)\det(C)$, $\det(Q^{-1}) = 1/\det(Q)$, and the fact that the determinant of a diagonal matrix is the product of its diagonal:

$$\det(A) = \det(Q\Lambda Q^{-1}) = \det(Q)\,\det(\Lambda)\,\det(Q^{-1}) = \det(Q)\left(\prod_{i=1}^{n}\lambda_i\right)\frac{1}{\det(Q)},$$

so

$$\boxed{\det(A) = \prod_{i=1}^{n}\lambda_i}$$

In particular, **if one of the eigenvalues is zero, the determinant is zero.**

Connecting to the meaning of the determinant. $|\det(A)|$ is the factor by which A scales areas. In the eigenbasis A stretches each axis by λ_i, so a unit square becomes a box with sides $\lambda_1, \ldots, \lambda_n$ and area $\prod_i \lambda_i$; the rotations Q^{-1} and Q undo each other's effect on area.

For the filter. $\det(S) = 2 \cdot 2 - 1 \cdot 1 = 3 = \lambda_1 \lambda_2 = 3 \cdot 1$. ✓ The picture triples in area. Check it on Figure 1: the logo's area goes from π to 3π (an oval with half-axes 3 and 1), and the unit square with sides e_1, e_2 becomes a parallelogram with sides $(2,1)$ and $(1,2)$, whose area is $|2 \cdot 2 - 1 \cdot 1| = 3$. For the practice filter, $\det(M) = 3 \cdot 0 - 2 \cdot 2 = -4 = 4 \cdot (-1)$. ✓ The area grows by $|-4| = 4$; the minus sign comes from the flip in Figure 3.

Check yourself. The filter is applied twice. By what factor has the area of the photo grown?
Answer. $3 \times 3 = 9$. (Section 7.3 confirms this: $\det(S^2) = 3^2 \cdot 1^2 = 9$.)

7.2 Inverse: the undo button

Claim: A^{-1} has the same eigenvectors as A, with eigenvalues $1/\lambda_i$.

Assume A is invertible, so all eigenvalues are nonzero ($\det(A) \neq 0$). Start from the eigenvalue equation and multiply both sides by A^{-1}:

$$Aq = \lambda q \quad \Longrightarrow \quad A^{-1}Aq = \lambda A^{-1}q \quad \Longrightarrow \quad q = \lambda A^{-1}q \quad \Longrightarrow \quad A^{-1}q = \frac{1}{\lambda}\, q.$$

So q is an eigenvector of A^{-1} with eigenvalue $1/\lambda$. In matrix form,

$$A^{-1} = (Q\Lambda Q^{-1})^{-1} = Q\,\Lambda^{-1}Q^{-1}, \qquad \Lambda^{-1} = \operatorname{diag}(1/\lambda_1, \ldots, 1/\lambda_n).$$

For the filter. Undoing "stretch the diagonal by 3" means "shrink the diagonal by $\frac{1}{3}$", along the same directions:

$$S^{-1} = Q \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & 1 \end{bmatrix} Q^T = \frac{1}{2} \begin{bmatrix} \frac{1}{3}+1 & \frac{1}{3}-1 \\ \frac{1}{3}-1 & \frac{1}{3}+1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

(The entry pattern $\frac{1}{2}(\lambda_1 + \lambda_2)$ on the diagonal and $\frac{1}{2}(\lambda_1 - \lambda_2)$ off it comes from multiplying out $Q\Lambda Q^T$ with this Q.) Check: the red pixel went to $(2,1)$, and $S^{-1}(2,1)^T = \frac{1}{3}(4-1,\, -2+2)^T = (1,0)^T$. The undo button works. ✓

Check yourself. Can the practice filter M be undone? If so, what are the eigenvalues of M^{-1}, and what is $M^{-1}(8,4)^T$?
Answer. Yes: neither eigenvalue is zero. M^{-1} has the same eigenvectors with eigenvalues $\frac{1}{4}$ and $\frac{1}{-1} = -1$. Undoing a flip is flipping again. Since $(8,4)^T = 4(2,1)^T$ lies along q_1, $M^{-1}(8,4)^T = \frac{1}{4}(8,4)^T = (2,1)^T$. (Check: $M^{-1} = -\frac{1}{4}\left[\begin{smallmatrix} 0 & -2 \\ -2 & 3 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 & 1/2 \\ 1/2 & -3/4 \end{smallmatrix}\right]$ gives $(0+2,\ 4-3)^T = (2,1)^T$. ✓)

7.3 Powers: tapping the filter again and again

Applying A twice:

$$A^2 = AA = Q\Lambda \underbrace{Q^{-1}Q}_{I} \Lambda Q^{-1} = Q\Lambda^2 Q^{-1},$$

and repeating the same cancellation,

$$\boxed{A^k = Q\,\Lambda^k\,Q^{-1}, \qquad \Lambda^k = \operatorname{diag}(\lambda_1^k, \ldots, \lambda_n^k)}$$

(The slides write A^n; we use k for the power to keep n for the size of the matrix.)

Connecting to the linear-transformation picture. Applying the filter k times means going round the loop of Figure 4 k times. Each "rotate back" (Q) is immediately undone by the next "rotate" (Q^{-1}), so a developer can rotate once, stretch k times, and rotate back once. Stretching k times by 3 is stretching by 3^k.

For the filter.

$$S^k = Q \begin{bmatrix} 3^k & 0 \\ 0 & 1 \end{bmatrix} Q^T = \frac{1}{2} \begin{bmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{bmatrix}.$$

Five taps give $S^5 = \frac{1}{2}\left[\begin{smallmatrix} 244 & 242 \\ 242 & 244 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 122 & 121 \\ 121 & 122 \end{smallmatrix}\right]$ with no multiplication of 2×2 matrices at all. (Check by brute force: $S^2 = \left[\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right]$, $S^4 = \left[\begin{smallmatrix} 41 & 40 \\ 40 & 41 \end{smallmatrix}\right]$, $S^5 = S^4 S = \left[\begin{smallmatrix} 122 & 121 \\ 121 & 122 \end{smallmatrix}\right]$. ✓)

What happens to the logo? Its long axis grows like 3^k while its short axis stays at 1, so the oval becomes a needle along the diagonal (Figure 5). The red pixel goes to $S^k(1,0)^T = \frac{1}{2}(3^k + 1,\ 3^k - 1)^T$, whose two coordinates become almost equal: the pixel is dragged towards the diagonal. Repeated application pulls everything towards the eigenvector with the largest eigenvalue.

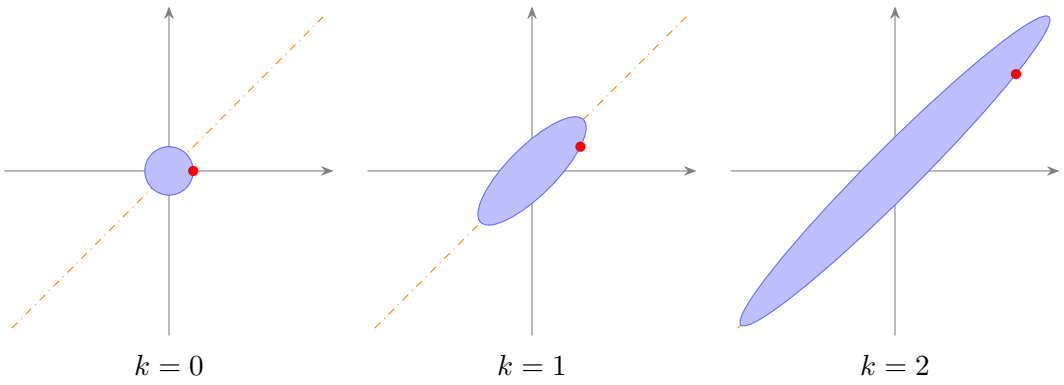


Figure 5: The logo after $k = 0, 1, 2$ taps of the filter (same scale in each panel). Half-axes are 3^k along the diagonal and 1 across it, so the logo becomes a needle. The red pixel, which started at $(1, 0)$, moves to $(2, 1)$ and then $(5, 4)$, closer and closer to the diagonal.

Check yourself. What does undoing five taps look like?
Answer. $S^{-5} = Q\,\mathrm{diag}(3^{-5}, 1)\,Q^T$: same directions, diagonal shrunk by $\frac{1}{243}$. Inverses and powers combine exactly as the eigenvalues do.

7.4 Zero eigenvalue: a filter that can never be undone

Suppose one eigenvalue is zero and q is an associated eigenvector. Then

$$Aq = \lambda q = 0, \qquad q \neq 0.$$

Recall that $Aq = q_{(1)}a_1 + q_{(2)}a_2 + \cdots + q_{(n)}a_n$ is a linear combination of the columns $a_1, \ldots, a_n$ of A, with weights the entries of q. So we have found a combination of the columns, with weights not all zero, that equals the zero vector. By the definition of linear independence, **the columns of A are not linearly independent**.

A "flatten" filter. Consider a filter that keeps the diagonal and removes the anti-diagonal completely:

$$F = Q \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} Q^T = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \qquad \lambda_1 = 1 \text{ along } (1,1), \quad \lambda_2 = 0 \text{ along } (-1,1).$$

Everything the lecture says about a zero eigenvalue shows up at once.

- **Dependent columns.** The two columns of F are identical, and the eigenvector $(-1, 1)$ is the recipe: $-1 \cdot \text{column } 1 + 1 \cdot \text{column } 2 = 0$.
- **Zero determinant.** $\det(F) = \frac{1}{4} - \frac{1}{4} = 0 = 1 \cdot 0$. The logo is flattened onto a line segment with no area (Figure 6).

- **No undo.** The pixels $(1,0)$ and $(0,1)$ both land on $(\frac{1}{2},\frac{1}{2})$. Given the flattened photo, there is no way to tell which pixel came from where, so no undo button can exist: F^{-1} does not exist.
- **Powers.** $F^k = Q\,\mathrm{diag}(1^k, 0^k)\,Q^T = F$: flattening twice is the same as flattening once.

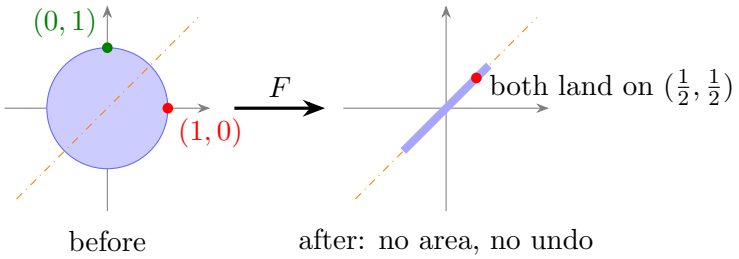


Figure 6: The flatten filter F has eigenvalue 0 across the diagonal. The logo becomes a segment, and two different pixels land on the same spot, so the original photo cannot be recovered.

Contrast with the practice filter. M has eigenvalues 4 and -1, neither zero, so its columns $(3,2)^T$ and $(2,0)^T$ are independent, $\det(M) = -4 \neq 0$, and it can be undone. A negative eigenvalue is harmless; only a zero eigenvalue loses information.

The four facts tie together: a zero eigenvalue $\iff$ dependent columns $\iff$ $\det(A) = 0$ $\iff$ A has no inverse (no undo).

Link to Notes 05. In least squares the matrix X^TX had to be invertible, and it failed when features were redundant (dependent columns). In eigenvalue language: redundant features give X^TX a zero eigenvalue.

8 Symmetric matrices: the spectral theorem

Every filter so far (S, M, F) was symmetric, and every one had perpendicular eigenvectors and a Q whose inverse was its transpose. This section explains why, and shows what a non-symmetric filter looks like.

8.1 The theorem

Spectral theorem. For a symmetric matrix $A \in \mathbf{R}^{n\times n}$, there exists an orthonormal basis of $\mathbf{R}^n$ consisting of eigenvectors of A, and each eigenvalue is real.

Orthonormal means the eigenvectors are mutually perpendicular and each has length one: $q_i^T q_j = 0$ for $i \neq j$ and $q_i^T q_i = 1$. Stacked as columns of Q, this says $Q^TQ = I$, that is,

$$Q^{-1} = Q^T \qquad (Q \text{ is an orthonormal matrix}).$$

8.2 Why $Q^{-1} = Q^T$ fits with symmetry

Start from the decomposition and transpose it, using $(BCD)^T = D^TC^TB^T$ and $\Lambda^T = \Lambda$:

$$A = Q\Lambda Q^{-1}, \qquad A^T = (Q^{-1})^T \Lambda Q^T.$$

Since $A = A^T$, these two expressions describe the same matrix. Choosing $Q^{-1} = Q^T$ makes them identical term by term: then $(Q^{-1})^T = Q$ and the second line becomes $Q\Lambda Q^T$ as well. The spectral theorem guarantees that this choice is always available for a symmetric matrix. So for symmetric A:

$$\boxed{A = Q\,\Lambda\,Q^T}$$

For the filter. S is symmetric; its eigenvalues 3 and 1 are real; its eigenvectors are orthonormal: $q_1^T q_2 = \frac{1}{2}(-1+1) = 0$ and $q_1^T q_1 = q_2^T q_2 = \frac{1}{2}(1+1) = 1$. Geometrically, an orthonormal Q is a rotation (or a reflection): it turns the picture without stretching or skewing it. So *every symmetric filter is "rotate, stretch along perpendicular axes, rotate back"*, and the logo always becomes a clean oval with perpendicular axes. The practice filter fits the same pattern: $q_1^T q_2 = \frac{1}{5}(2-2) = 0$, and its negative eigenvalue just makes the "stretch" step a flip along q_2.

8.3 A badly behaved filter

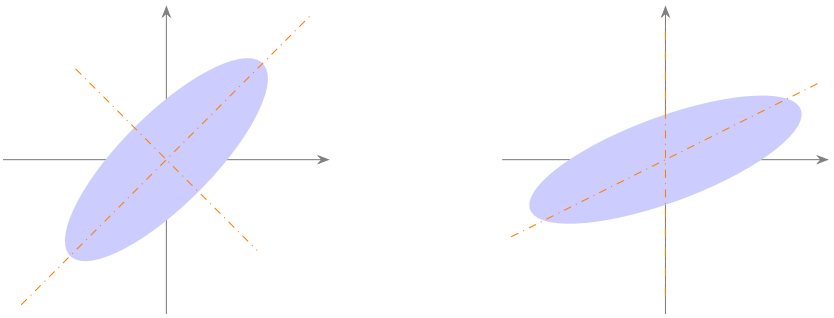
Compare a lopsided filter with the *same* eigenvalues as S:

$$L = \begin{bmatrix} 3 & 0 \\ 1 & 1 \end{bmatrix}, \qquad \det(L - \lambda I) = (3-\lambda)(1-\lambda) \;\Rightarrow\; \lambda_1 = 3,\ \lambda_2 = 1.$$

Running the recipe: for $\lambda_1 = 3$, $(L - 3I)v = 0$ gives $v_1 = 2v_2$, so $q_1 \propto (2, 1)$; for $\lambda_2 = 1$, $(L - I)v = 0$ gives $v_1 = 0$, so $q_2 \propto (0, 1)$. These are *not* perpendicular: $(2, 1) \cdot (0, 1) = 1 \neq 0$. The decomposition still exists,

$$L = Q\Lambda Q^{-1}, \qquad Q = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}, \quad Q^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix},$$

(check: $Q\Lambda Q^{-1} = \frac{1}{2}\left[\begin{smallmatrix} 6 & 0 \\ 3 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 0 \\ -1 & 2 \end{smallmatrix}\right] = \frac{1}{2}\left[\begin{smallmatrix} 6 & 0 \\ 2 & 2 \end{smallmatrix}\right] = L$ ✓), but now $Q^{-1} \neq Q^T$. The change of basis is no longer a rotation: it is a skew. The stretch directions in Figure 7 are slanted relative to each other, and the axes of the output oval are not the eigenvector directions.



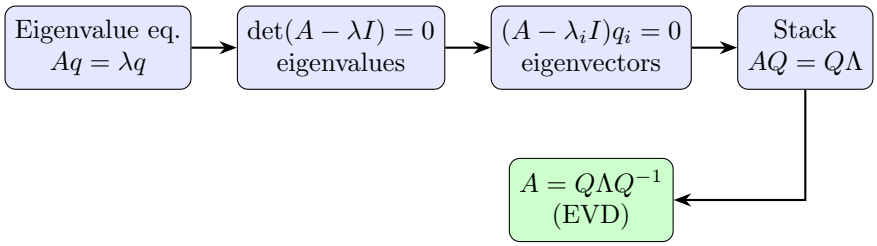
symmetric S: perpendicular axes lopsided L: slanted stretch directions

Figure 7: The logo under two filters with the same eigenvalues (3 and 1). Dash-dotted lines are the eigenvector directions. For the symmetric S they are perpendicular and form the oval's axes; for L they are slanted and do not.

Summary. For a symmetric matrix, the eigenvectors are orthonormal and the eigenvalues are real.

Check yourself. The "rotate by 90°" filter is $R = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, which is not symmetric. Find its characteristic polynomial. Does any pixel direction stay on its own line?
Answer. $p(\lambda) = \lambda^2 + 1$, which has no real roots. So R has no real eigenvalues and no real eigenvectors: every direction is turned by 90°, and none stays on its line. The spectral theorem's promise that "each eigenvalue is real" is a real guarantee: non-symmetric matrices can fail it.

9 Summary map



Idea	**Key formula or fact**	**In the photo filter S**
Eigenvector, eigenvalue	$Aq = \lambda q$, $q \neq 0$, A square	Diagonal stretched by 3; anti-diagonal unchanged (1)
Computing them	$p(\lambda) = \det(A - \lambda I) = 0$, degree n; then $(A - \lambda I)q = 0$	$\lambda^2 - 4\lambda + 3$; $q_1 = \frac{1}{\sqrt{2}}(1,1)$, $q_2 = \frac{1}{\sqrt{2}}(-1,1)$
Vocabulary	Eigenspectrum = set of eigenvalues; eigenspace = span of eigenvectors	Spectrum $\{3,1\}$; eigenspace of 3 = the diagonal
Eigendecomposition	$AQ = Q\Lambda \Rightarrow A = Q\Lambda Q^{-1}$	Needs independent eigenvectors (Q invertible)
Interpretation	Q^{-1}: change of basis; Λ: scaling; Q: change back	Rotate $-45°$, stretch sideways by 3, rotate $+45°$
Determinant	$\det(A) = \prod_i \lambda_i$	Area $\times 3$ (logo: $\pi \to 3\pi$)
Inverse	$A^{-1} = Q\Lambda^{-1}Q^{-1}$, eigenvalues $1/\lambda_i$	Undo button: shrink diagonal by $\frac{1}{3}$
Powers	$A^k = Q\Lambda^k Q^{-1}$	k taps: diagonal $\times 3^k$, logo becomes a needle
Zero eigenvalue	$Aq = 0$ with $q \neq 0 \Rightarrow$ dependent columns, $\det = 0$, no inverse	Flatten filter F: identical columns, no area, no undo
Symmetric matrices	$A = Q\Lambda Q^T$, real eigenvalues, orthonormal eigenvectors	Q is a rotation; lopsided L skews instead
Practice filter	$M = \left[\begin{smallmatrix} 3 & 2 \\ 2 & 0 \end{smallmatrix}\right]$: $\lambda = 4, -1$	$q_1 = \frac{1}{\sqrt{5}}(2,1)$, $q_2 = \frac{1}{\sqrt{5}}(1,-2)$; stretch and flip; $\det = -4$; invertible